

JavaScript Task

Introduction

- JavaScript is a web scripting language and it is used to change both the HTML and CSS content and mainly it is used to calculate, manipulate and validate data.
- The JavaScript can be placed in the HTML *head* or *body* or *both*.

JavaScript Basics

- JavaScript can reach out the HTML page by using different ways like `getElementById(id)`
e.g, if we have `<p id="pid">The paragraph text </P>`
- we can reach out to the p element by `document.getElementById('pid')`
- There are different methods of displaying in JavaScript.
 1. Writing into an HTML element by using `innerHTML` or `innerText`.
 2. Writing into the HTML output by using `document.write()`.
 3. Writing into an alert box by using `window.alert()`.
 4. Writing into the browser console by using `console.log()`.

Variables and Data Types

Variables are identifier and they:

- ▶ contain letters, dolar sign, underscore and digits,
- ▶ start with letters, underscores, or dolar sign(\$),
- ▶ are case sensitive(age and Age are different variables)
- ▶ reserved word can't be variable name(let, const we can't use them as variables)

Variables can declare by using:

- ▶ `let`(can be modified but not redeclared) like `let x` or multiple `let x, y, z = 9`.
- ▶ `const`(can't be modified and reassigned) like `const x = 0`.
- ▶ `var`(it is global and can be used every where and I understand it is not recommended it is the old way of declaring variable.
- ▶ `default`(also not recommended).

Data types

A JavaScript variable can hold 8 types of data:

Type	Description
String	A text of characters enclosed in quotes
Number	A number representing a mathematical value
Bigint	A number representing a large integer
Boolean	A data type representing true or false
Object	A collection of key-value pairs of data
Undefined	A primitive variable with no assigned value Undefined is a value automatically assigned to a variable that has been declared but has no value.

Null	A primitive value representing object absence
Symbol	A unique and primitive identifier

Operators

Operators are things used for Computations.

- **Arithmetic** to perform mathematical operations
 - ▶ Example:- Addition(+), Subtraction(-), multiplication(*), Exponentiation(**), Division(/), Remainder or modulus(%), Increment(++), and Decrement(--)
- **Assignment** used to assign values
 - ▶ Example :- =, +=, -=, *=, /= and others
- **Comparisons** used to compare different things
 - ▶ Example:- compare values(==), compare value and type(===), >, <, <=, >=, != and others.....
- **Logical** operators used to check two or more conditions or to check more than one things
 - ▶ Example:- And (&&), Or (||), Nor or Negation(!)

Control Statements and Loops

- **control statements** are used to control a program in different conditions like when something is true do something and when false do other thing
 - ▶ Example:- if, else and else if and switch(when we get case situations) statements.
- **Loops** are used to iterate some actions *like we write one program witch execute many times*
 - ▶ Example:- for loop, while loop and do while loop(when we want to execute the program at least once).

Functions

Functions are a reusable code block designed for a particular task and they execute only when they called.

```
Example:- function add(a, b) {
    return a + b;
} ; // Currently it does nothing just waiting till called

let result = add(5, 4); // when it called it accept the params and return the result.
```

Objects

Objects is variable used to store data with with key and values.

Example:- `let student = {id:"001", grade:3.5, department:"SE"}` and

- we can access values by using "." like `student.id` Or "[" like `student["id"]` . -
- we can modify and add the property like : `student.grade = 4` .or `student.age=22`